University of St.Gallen

PROF. DR. ALEXANDER BRAUN
ASSOCIATE PROFESSOR OF INSURANCE AND CAPITAL MARKETS

Stochastic Modelling in Finance and Insurance

Problem Set 2

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Instructions

Please submit your answers as a **text-based file** (PDF or Word) containing all required deliverables, such as numerical results, figures, and brief discussions. Add your properly annotated and executable **R code**. You are encouraged to use code snippets from the course and **Generative AI** for tutoring, coding, and writing. However, please remember that over-reliance on AI may compromise your learning effect. Finally, all work must be your own; copying from others is strictly prohibited.

Exercise 1: Importing and preparing the data

This problem set is based on the accompanying file "Asset_Classes.csv", which contains monthly total return time series for diversified portfolios of four different asset classes (stocks, treasuries, corporate bonds, and real estate) in the period from January 1993 to December 2012. It also contains the corresponding one-month risk-free interest rates (Rf).

Use the function `read.csv2()` to import the data into R and store it into a data frame. Check the structure of the data frame with the function `str()`. In case R has taken the data for character strings (chr), transform it into numeric data by means of `as.numeric()`.

Exercise 2: The Markowitz model

Estimate the input parameters for the Markowitz model (expected returns, standard deviations, and the covariance matrix of the returns) from the entire time series available for the asset class portfolios (i.e., January 1993–December 2012).

Then employ `solve.QP()` (quadprog package) to run a Markowitz optimization *with short-sale constraints* to determine the weights for efficient portfolios that combine the four asset classes (perspective of an investor on the 31st of December 2012).

Report: A plot of the efficient frontier, the minimum variance portfolio, and all four basic asset class portfolios in the μ-σ space.

Exercise 3: Optimal vs. Naïve Diversification

Now split the time series data into two subsamples: Use the 120-month period between January 1993 and December 2002 for parameter estimation and reserve the 120-month period between January 2003 and December 2012 as a test sample.

a) Reestimate your input parameters for the **Markowitz model** based on the estimation sample (January 1993 to December 2002). Run the optimization again and determine the portfolio weights of the corresponding efficient frontier *with short-sale constraints* (perspective of an investor on the 31st of December 2002).

b) Form the return time series of the **1/N strategy**, i.e., an equally weighted portfolio of all four asset classes, in the period January 1993 to December 2002. Compute sample mean and sample standard deviation of these "historical" returns.

   [*Hint*: Use the R commands `mean(.)` and `sd(.)` on the return vector.]

c) Find the efficient portfolio that has the same standard deviation as the 1/N portfolio.

   [*Hint*: The easiest way to do this is to determine the index position of the portfolio in the vector that contains the standard deviations of the efficient portfolios. This can be achieved with the following syntax: `which.min(abs())`. The argument of `abs()` is the difference between the standard deviation of the 1/N portfolio and the vector of standard deviations of the efficient portfolios.]

Report: A plot of the efficient frontier, the minimum variance portfolio, the 1/N portfolio, the efficient portfolio with the same standard deviation as the 1/N portfolio, and all asset class portfolios in the μ-σ space.

d) Compute the time series of "realized" returns for the 1/N portfolio and the *corresponding* efficient portfolio in the test period (January 2003 to December 2012). Use these return time series to determine the out-of-sample Sharpe Ratios, defined as the mean of the out-of-sample excess returns ($\hat{\mu}^e$) divided by their standard deviation ($\hat{\sigma}^e$):

$$Sharpe\ Ratio = \frac{\hat{\mu}^e}{\hat{\sigma}^e}$$

with

$$\hat{\mu}^e = \frac{1}{T}\sum_{t=1}^{T}(r_t^e) \qquad \hat{\sigma}^e = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}(r_t^e - \hat{\mu}^e)^2},$$

where $r_t^e = r_t - r_t^f$ is the out-of-sample excess return in month t ($t = 1$ is January 2003 and $t = T$ is December 2012, i.e., $T = 120$).

Report: The out-of-sample Sharpe Ratios of the 1/N portfolio and the *corresponding* efficient portfolio. What do you observe? Provide an explanation for your result.

General Hints:

- The returns and risk-free rates in the input file are reported as decimal figures. Thus, 0.0418, e.g., means 4.18%.
- You may opt for a step size of 0.0001 in the optimization as a compromise between precision and computation time.
- Use the contemporaneous risk-free interest rate for the calculation of the excess returns $r_t^e = r_t - r_t^f$. That is, for January 2003 use the risk-free rate in January 2003, for February 2003 use the risk-free in February 2003, and so forth.
- If `solve.QP(.)` fails, try min(MU)+0.0000000001 as the lower bound of the range.
- Instead of using the date format in the first column, you may simply reference rows in your data frame by their index.
    - January 1993 to December 2002 comprises rows 1 to 120.
    - January 2003 to December 2012 comprises rows 121 to 240.